

DDR基础课程

第四节 - 初步了解DRAM Jedec Spec 2

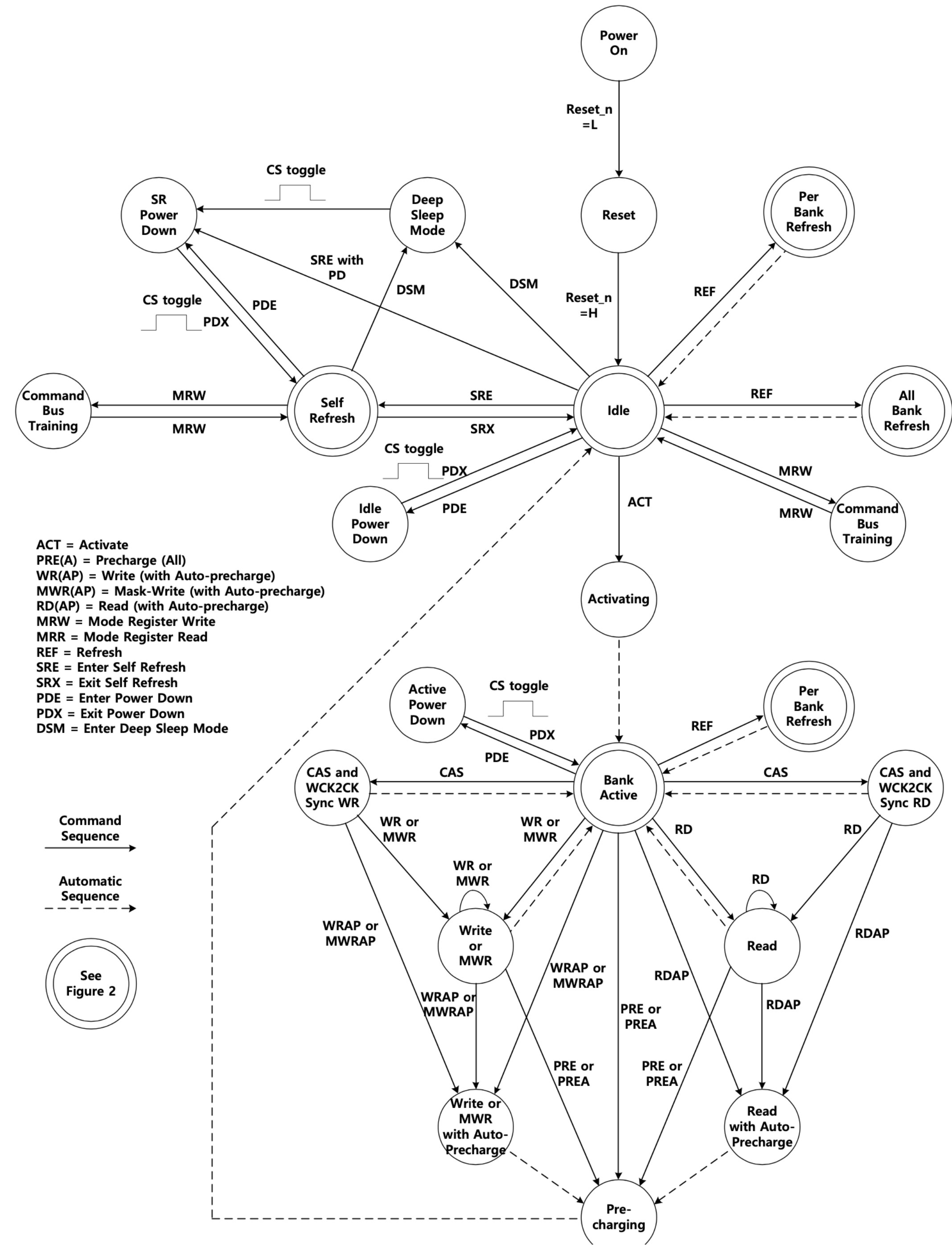
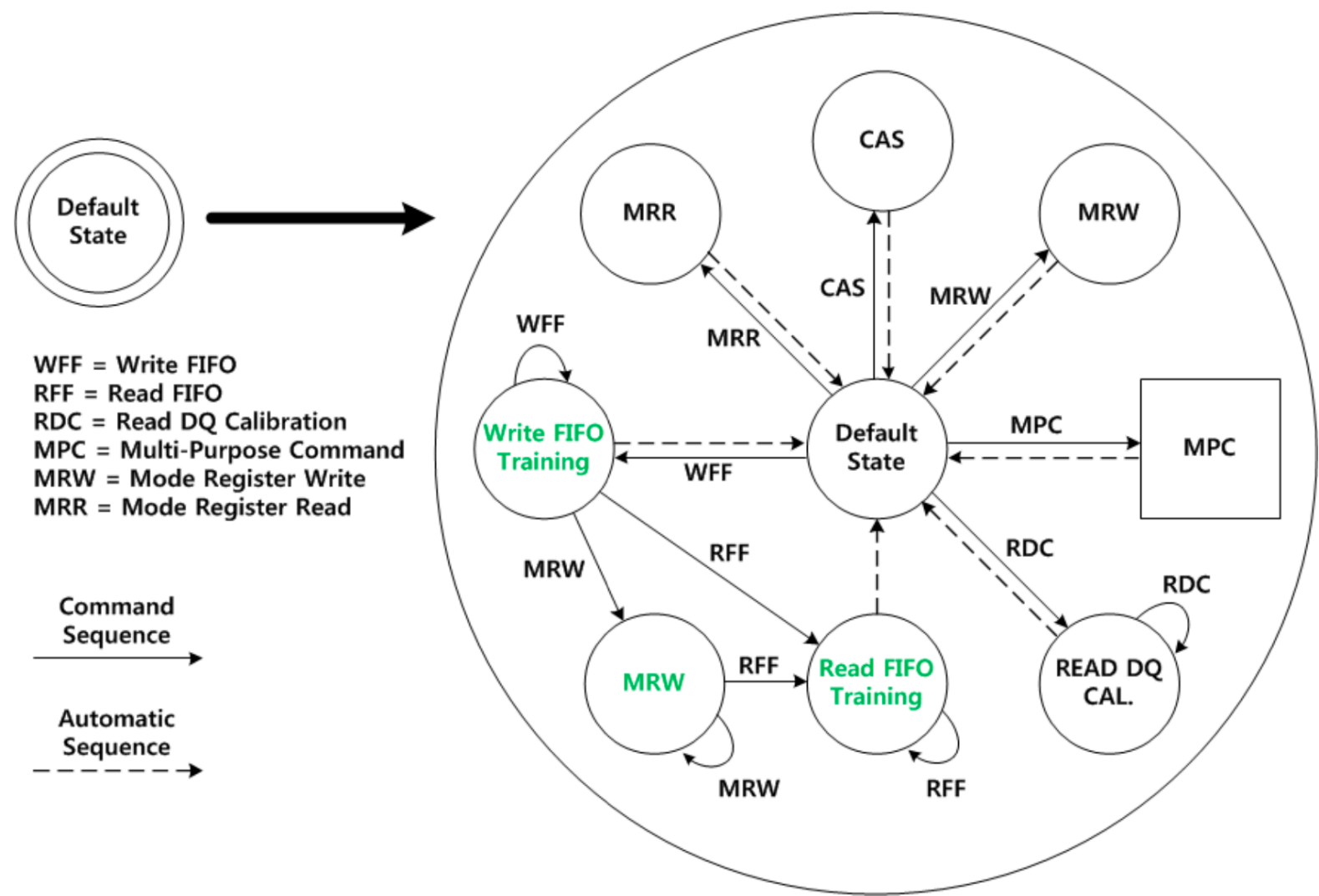
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关注的问题

- 命令有哪些？为什么需要这些命令？
- 命令与状态的关系是什么？
- 命令的格式是什么样的？
- 命令与数据的波形是什么样的？
- 从用户的角度看如何读取一个数据回来

状态机



命令

- 真值表

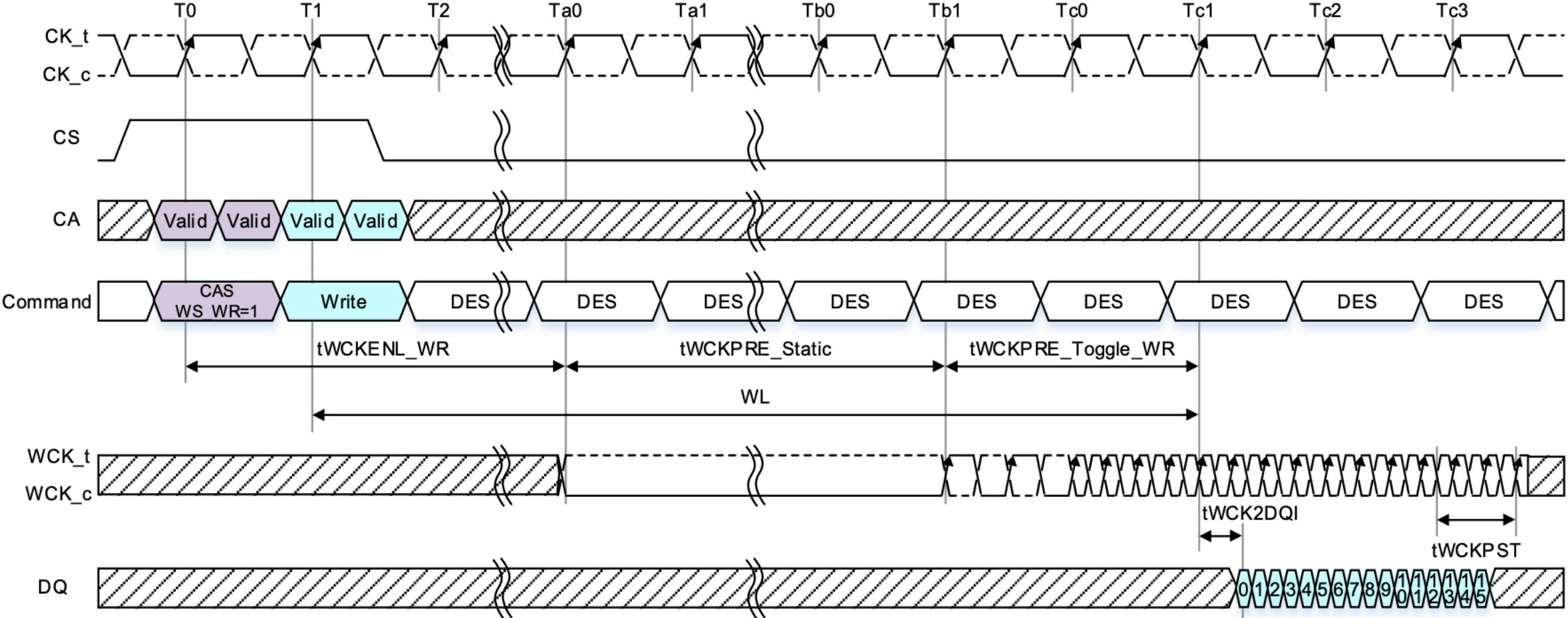


Table 197 — Command Truth Table

SDRAM COMMAND	BK ORG (BG, 16B, 8B)	SDR CMD PIN	DDR COMMAND PINS							CK_t edge	Notes
		CS	CA0	CA1	CA2	CA3	CA4	CA5	CA6		
DESELECT (DES)	Any	L	X	X	X	X	X	X	X	R1	1, 2
		X	X	X	X	X	X	X	X	F1	
NO OPERATION (NOP)	Any	H	L	L	L	L	L	L	L	R1	1, 2
		X	X	X	X	X	X	X	X	F1	
POWER DOWN ENTRY (PDE)	Any	H	L	L	L	L	L	L	H	R1	1, 2, 12
		L	X	X	X	X	X	X	X	R2	
ACTIVATE-1 (ACT-1)	Any	H	H	H	H	R14	R15	R16	R17	R1	1, 2, 3, 4
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	R11	R12	R13	F1	
	8B		BA0	BA1	BA2	V					1, 2, 4
	Any	H	H	H	L	R7	R8	R9	R10	R1	
		X	R0	R1	R2	R3	R4	R5	R6	F1	
PRECHARGE (PRE) (Per Bank, All Banks)	Any	H	L	L	L	H	H	H	H	R1	1, 2, 3, 5
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	V	V	AB	F1	
	8B		BA0	BA1	BA2	V					1, 2, 3, 5, 14
	Any	H	L	L	L	H	H	H	L	R1	
	BG		BA0	BA1	BG0	BG1	SB0	V			
REFRESH (REF) (Per Bank, All Banks)	16B	X	BA0	BA1	BA2	RFM	SB0	V	AB	F1	1, 2, 3, 6, 7
	8B		BA0	BA1	BA2		V	SB1			
	Any	H	L	H	L	C0	C3	C4	C5	R1	
MASK WRITE (MWR)	BG		BA0	BA1	BG0	BG1					1, 2, 3, 6, 7, 8
	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	
	8B		BA0	BA1	BA2	V					
WRITE (WR16 or WR)	Any	H	L	H	H	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 8
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	
	8B		BA0	BA1	BA2	V					1, 2, 3, 6, 7, 8
	BG/16B	H	L	L	H	L	C3	C4	C5	R1	
	BG		BA0	BA1	BG0	BG1					
WRITE32 (WR32)	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	
	8B		BA0	BA1	BA2	B4					
READ (RD16 or RD)	Any	H	H	L	L	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 10, 11
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	
	8B		BA0	BA1	BA2	B4					1, 2, 3, 6, 7, 10, 11
	BG/16B	H	H	L	H	C0	C3	C4	C5	R1	
	BG		BA0	BA1	BG0	BG1					
READ32 (RD32)	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	

命令

- 真值表

Table 198 — Allowable CAS command operand(s) combination

Operation Mode	CAS Command Operand(s)						
	WS_WR	WS_RD	WS_FS	DC0-3	WRX	WXSA	WXSBB3
WCK2CK SYNC (WR)	H	L	L	LLLL	L	L	L
WCK2CK SYNC (WR) + Write Data Copy	H	L	L	VVVV	L	L	L
WCK2CK SYNC (WR) + Write X	H	L	L	LLLL	H	V	V
WCK2CK SYNC (RD)	L	H	L	LLLL	L	L	V
WCK2CK FAST SYNC	L	L	H	LLLL	L	L	L
WCK2CK SYNC OFF	H	H	H	LLLL	L	L	L
Write Data Copy	L	L	L	VVVV	L	L	L
Write X	L	L	L	LLLL	H	V	V
Non-Zero Burst Start Address for RD	L	L	L	LLLL	L	L	H
WCK SUSPEND	H	L	H	LLLL	L	L	L
Illegal		Other Operands Combination					
NOTE 1 "V" means "H" or "L" (a defined logic level). NOTE 2 All CAS command operands are non-sticky. NOTE 3 Write Data Copy function requires DC0-3 operands not to be all-zero. NOTE 4 CAS-WCK_SUSPEND command is valid only when MR0 OP[2]=1B (Enhanced WCK Always On mode supported) and MR18 OP[4]= 1B (WCK Always On mode enabled)							

Table 197 — Command Truth Table (cont'd)

SDRAM COMMAND	BK ORG (BG, 16B, 8B)	SDR CMD PIN	DDR COMMAND PINS							CK_t edge	Notes
		CS	CA0	CA1	CA2	CA3	CA4	CA5	CA6		
CAS	Any	H	L	L	H	H	WS_WR	WS_RD	WS_FS	R1	1, 2, 7, 13
		X	DC0	DC1	DC2	DC3	WRX	WXSA	WXSB/B3	F1	
MULTI PURPOSE COMMAND (MPC)	Any	H	L	L	L	L	H	H	OP7	R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	F1	
SELF REFRESH ENTRY (SRE)	Any	H	L	L	L	H	L	H	H	R1	1, 2, 9
		X	V	V	V	V	V	DSM	PD	F1	
SELF REFRESH EXIT (SRX)	Any	H	L	L	L	H	L	H	L	R1	1, 2
		X	V	V	V	V	V	V	V	F1	
MODE REGISTER WRITE-1 (MRW-1)	Any	H	L	L	L	H	H	L	H	R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	F1	
MODE REGISTER WRITE-2 (MRW-2)	Any	H	L	L	L	H	L	L	OP7	R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	F1	
MODE REGISTER READ (MRR)	Any	H	L	L	L	H	H	L	L	R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	F1	
WRITE FIFO (WFF)	Any	H	L	L	L	L	L	H	H	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
READ FIFO (RFF)	Any	H	L	L	L	L	L	H	L	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
READ DQ CALIBRATION (RDC)	Any	H	L	L	L	L	H	L	H	R1	1, 2
		X	L	L	L	L	L	L	L	F1	
NOTE 1 LPDDR5 commands are one clock cycles long and defined by the states of CS at the rising edge (R1) of clock and CA[6:0] at the rising edge (R1) and the falling edge (F1) of clock. Note that some operations such as ACTIVATE, MODE REGISTER WRITE or WCK2CK SYNC (WS_RD, WS_WR, WS_FS) require two commands to initiate.											
NOTE 2 "V" means "H" or "L" (a defined logic level). "X" means don't care in which case CK_t, CK_c and CA[6:0] can be floated.											
NOTE 3 Bank Group Address BG[1:0] and Bank Address BA[1:0] determine which bank is to be operated upon.											
NOTE 4 An ACTIVATE-1 command must be followed by an ACTIVATE-2 command. Only CAS, WRITE, MASK WRITE, READ, PRECHARGE (to a different bank) commands can be issued between ACTIVATE-1 and ACTIVATE-2 commands. The maximum timing gap between ACTIVATE-1 and ACTIVATE-2 commands are 8 clock cycles. An ACTIVATE-1 command must be issued first before issuing an ACTIVATE-2 command. Once an ACTIVATE-1 command is issued, an ACTIVATE-2 command must be issued before issuing another ACTIVATE-1 command.											
NOTE 5 AB "HIGH" during PRECHARGE or REFRESH command indicates that command must be applied to all banks and bank address is a don't care.											

上节课后问题解答

- 课后问题1: Apple M1芯片的DDR通道数为8? 其对应的DRAM类型为LPDDR4X? 最高速率为4266Mbps?最高总带宽为 $8 \times 16\text{bit} \times 4266\text{Mbps} = 68,256\text{MB/s}$?
- 课后问题2: “如果总带宽不足, 可以通过将Rank数翻倍, 从而将带宽翻倍”。以上说法是正确的吗? 错误。无论有多少个Rank, 都是share一个channel的, 那么只能提升容量, 提升不了带宽。如果要提升带宽, 则应当考虑提升速率或者通道数量。
- 课后问题3: “data rate指的是CK的频率的两倍”。以上说法是正确的吗? 错误。data rate指的是dq的频率, 也就是WCK频率的两倍。
- 课后问题4: “data rate指的是WCK的频率”。以上说法是正确的吗? 错误。data rate指的是dq的频率, 也就是WCK频率的两倍。
- 课后问题5: DMI信号应该与以下哪个时钟对齐? A. CK; B. WCK B。DMI信号与DQ信号一样, 为数据信号, 因此要与WCK对齐。
- 课后问题6: ZQ信号外接的电阻大小为240(DDR & LPDDR)/120(GDDR)欧?

课后问题

- 课后问题1: “DDR指的是数据是双沿的。但是在LPDDR5中的命令也是上下沿采样的”。以上说法是正确的吗?
- 课后问题2: “Deselect 与 NOP对DRAM产生的作用是相同的, 因此两个命令可以合二为一, 或者说其中一个是冗余的”。以上说法是正确的吗?
- 课后问题3: “Precharge命令与Refresh命令均有all bank和per bank两种模式”。以上说法是正确的吗?
- 课后问题4: LPDDR5中CAS命令中与WCK2CK Sync有关的命令有___个? 分别为___? 他们的含义分别是___?
- 课后问题5: 在发送Read命令前, 必须保证目标row处于()状态, 且必须保证WCK2CK处于(Sync/Unsync)状态?
- 课后问题6: 要使得WCK2CK处于Sync状态, 有哪些方法? 要使得WCK2CK处于Unsync状态, 有哪些方法?
- 课后问题7: 发送Activate命令的前提条件是此时DRAM处于___状态? 如果不处于此状态, 如何使得DRAM进入此状态?

下节预告

第五节 - 初步了解DRAM Jedec Spec 3 从用户的角度看读写

- 从用户的角度看如何读取一个数据回来
- 从用户的角度看如何写一个数据进去

谢谢